

TEJASWIN KRISHNA

Senior Researcher, Energy System Materials Lab,
Department of Advanced Materials Engineering,
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CAREER OBJECTIVE

To excel in the field of welding engineering by acquiring and improving the skill sets that will enable me to fulfil my personal and professional objectives.

RESEARCH EXPERIENCE

Senior Researcher September 2024- Present
Dong-Eui University, Busan, Republic of Korea

- Developing machine learning algorithms for quality prediction of RSW welds
- Extraction of features from STL files and images using different machine learning models to perform regression analysis.

Graduate Research assistant March 2022- August 2024
Dong-Eui University, Busan, Republic of Korea

- Leading project focusing on optimizing resistance spot welding sequences for automotive components, primarily based on finite element analysis (FEA) and machine learning
- Completed 1 project on developing an optimal electrode geometry for Resistance spot welding of dissimilar thickness Aluminum alloys for short flange applications.
- Experience with finite element analysis (FEA) tools such as SORPAS® and SIMUFACT WELDING.
- Performed in-situ analysis of resistance spot welding using high-speed camera and infrared thermography.

EDUCATION

Dong-Eui University, Busan, Republic of Korea (March 2022- August 2024)
M.S from Advanced Materials engineering department, GRADE: 4.50/4.50 GPA
specializing in Welding, and joining technology
Academic advisor: Prof. Yeong-Do Park

APJ Abdul Kalam Technological University, Kerala, India (August 2016-September 2020)
Bachelor of technology in Metallurgy GRADE: 6.61/10.0 GPA
Academic advisor: Mebin T Kuruvila

RESEARCH PROJECT HANDLED DURING ACADEMICS

MAJOR CONTRIBUTIONS

1. Optimization of Resistance Spot Welding Sequence for Different Automotive Components

(Project No: T101600323030001)
(FUNDED BY HYUNDAI MOTOR CORPORATION, KOREA) (September 2023- Present)

- Analysis of underlying factors contributing to deformation during welding automotive components.
- Generating welding sequences with mathematical models and machine learning
- Verification of the sequences with welding simulation

• Validation of the generated sequences by real welding
2. In-line Quality Inspection Using 3D Scanned Indentation of Resistance Spot Welding
(In collaboration with TESONE, KOREA) (March 2023-Present)
• Actively contributed as part of the team in developing machine learning algorithms and equations to improve inspection accuracy. • Prepared resistance spot welding of various automotive steel weld grades to enhance the technology's database.
3. Development of optimal electrode geometry for the Resistance Spot Welding of Aluminum alloys for short flange applications
(Project No: T012600222060008)
(FUNDED BY HYUNDAI MOTOR COMPANY, KOREA) (May 2022-March 2023)
• Designing of a new electrode geometry which improved the heat balance, nugget growth rate and electrode life. • Optimization of welding conditions for the Resistance spot welding of dissimilar thickness Aluminum alloys • Development of mathematical correlation to locate heat balance in dissimilar thickness combination. • Comparison with conventional electrode geometries on electrode degradation
MINOR CONTRIBUTIONS
1. Finite Element Analysis Analysis of temperature distribution, tensile stress and strain at the locations susceptible to LME cracks using finite element modelling
• Effect of decarburizing layer on Liquid Metal Embrittlement in Advanced High Strength Steels (In collaboration with Korea Institute of Material Science (KIMS)) • Investigation and Prevention of Liquid Metal Embrittlement Cracking in Uncoated Transformation-Induced Plasticity Steel through Resistance Spot Welding (In collaboration with Korea Institute of Industrial Technology (KITECH)) • Development of smart factory-based future vehicle body module welding and mechanical fastening quality prediction system (FUNDED BY GRAND ICT, KOREA)
PUBLICATIONS AND CONFERENCE PROCEEDINGS
PUBLICATIONS

1. Jae Won Kim, Sunusi Marwana Manladan, Kaisar Mahmud, Woosung Jin, Tejaswin Krishna, Changwook Ji, Dae-Geun Nam and Yeong-Do Park “Liquid Metal Embrittlement Cracking in Uncoated Transformation-Induced Plasticity Steel during Consecutive Resistance Spot Welding” — Metals https://doi.org/10.3390/met13111826 2. Md Abdul Karim, Sunusi Marwana Manladan, Hasan M.M. Afroz, Woosung Jin, Tejaswin Krishna, Changwook Ji, Duck Bong Kim, Yeong-Do Park “Critical effect of heat input on joint quality in resistance element welding of Al and steel” — Journal of Manufacturing Processes https://doi.org/10.1016/j.jmapro.2023.04.005 3. A review on “Manufacturing of a bimetallic structure of Stainless Steel and Mild Steel through Wire Arc Additive Manufacturing” — IRJET Journal	
CONFERENCE PROCEEDINGS (SELECTED FEW)	
1 Tejaswin Krishna, et. al. “Optimization of electrode geometry in aluminum resistance spot welding of dissimilar thickness combination” — Presented at IIW ANNUAL ASSEMBLY MARINA BAY SANDS CONVENTION CENTRE, SINGAPORE 2023 2 Tejaswin Krishna, et. al. “Fusion Assisted Mechanical Fastening (EASW and REW) technology for dissimilar materials joining” — Presented at INTERNATIONAL WELDING & JOINING CONFERENCE - KOREA 2022 3 Tejaswin Krishna, et. al. “Effect of electrode geometry on aluminum resistance spot welding of dissimilar material thickness combinations” Presented at KOREAN WELDING AND JOINING SOCIETY CONFERENCE SPRING, KOREA 2023	
SKILLS	
• Welding	Operation of Resistance spot welding machines
• Robot Programming	RSW Robotic Controller
• Material characterization	OM, SEM, EBSD, EPMA
• Material testing	Tensile testing, Hardness testing
• Process visualization	High-Speed and Infrared (IR) imaging
• Finite Element Analysis	SORPAS®, SIMUFACT WELDING, ANSYS (elementary proficiency)
• Programming languages	C, C++, Python (elementary proficiency)
• English Language Proficiency	Listening – 8.5, Reading – 8.5, Writing – 6.0, Speaking – 7.5 Overall score – 7.5
REFEREES	
Prof. Yeong-Do Park	Dr. Sunusi M. Manladan
Department of Advanced Materials Engineering, Dong Eui University Busan, Republic of Korea (Zip: 614-714) Email: ypark@deu.ac.kr Ph: +82-10-6429-1860 (Mobile)	Senior Lecturer, Department of Mechanical engineering, Bayero University, Kano, Federal Republic of Nigeria (Zip: 700006) Email: smanladan.mec@buk.edu.ng Ph: +234 803 813 9932

PERSONAL DETAILS	
Date of Birth	27 th December 1997
Nationality	Indian
Sex/Gender	Male
Marital status	Single
COMMUNICATION ADDRESS	
CURRENT ADDRESS	PERMANENT ADDRESS
Energy System Materials Lab, Department of Advanced Materials Engineering, Dong Eui University Busan, Republic of Korea (Zip: 614-714)	Sree Bindu Nivas, P.O. Chalappuram, Kozhikode, Kerala, India (Zip 673-002)
DECLARATION	
I solemnly declare that the above-mentioned information's are true to the best of my knowledge	
Tejaswin Krishna	14 th January 2025